Sensitivity of the Philadelphia's beverage tax effect to violations of parallel trends: A Bayesian sensitivity analysis

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Abstract

Difference-in-differences (DiD) designs are widely used to evaluate the causal effects of public policies, yet their validity depends critically on the parallel trends assumption. In policy settings such as excise taxes on consumer goods, changes in consumer purchasing behavior, retailer responses, and other time-varying factors make parallel trends difficult to justify, complicating interpretation of estimated effects. Using Philadelphia's sweetened beverage tax as a case study, with Baltimore as a control, we propose a Bayesian DiD framework that introduces a sensitivity parameter to represent possible deviations from the parallel trends assumption. We place an autoregressive AR(1) prior with a mean-shift component on these deviations, allowing counterfactual trends to evolve gradually while accommodating persistent departures from parallel trajectories. To assess robustness, we consider multiple prior specification strategies, including fixed hyperparameters, fully Bayesian priors, and empirical Bayes approaches calibrated from pre-treatment data, and compare posterior treatment effect estimates across these configurations. We further conduct a tipping point analysis that quantifies how large violations of parallel trends would need to be to attenuate the estimated policy effect toward the null, translating sensitivity parameters into interpretable changes in beverage sales. Together, these analyses show how Bayesian sensitivity analysis can clarify the robustness of policy conclusions in realistic DiD applications where parallel trends are uncertain, providing a practical framework and transparent guidance for interpreting causal effects in applied policy evaluation.

Keywords: Difference-in-differences · Bayesian sensitivity analysis · Causal inference · Policy evaluation

1 Introduction

To evaluate the causal impact of public policy, practitioners commonly use DiD designs, which compare outcome trends between treated and control groups while adjusting for time-invariant differences [2]. A key identifying assumption in DiD is that, in the absence of treatment, the outcomes of the two groups would have followed parallel trends. This assumption is often difficult to justify in real-world policy evaluations, where complex pre-treatment dynamics, time-varying confounding, or unobserved effect modifiers may violate the parallel trends condition [22]. For instance, a DiD analysis of Philadelphia's sweetened beverage tax using Baltimore as a control found a 51% decline in city-level beverage sales following the tax [18]. However, such estimates may be sensitive to deviations from the assumed counterfactual trend, since Philadelphia and Baltimore may differ in unobserved factors such as consumer characteristics and

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purchasing behavior or broader regional economic conditions that can influence beverage sales trajectories over time.

A common approach for evaluating the plausibility of parallel trends is pre-trends "testing", which checks whether treated and control groups followed similar outcome trajectories before treatment. While widely used, such tests are underpowered in short panels or with noisy outcomes and do not guarantee that trends would have remained parallel post-treatment. Further, as noted by [19], pre-tests can introduce selective inference when only favorable results are reported. Moreover, failing to detect a pre-treatment difference does not confirm the assumption holds after treatment [5]. To address potential deviations more directly, researchers have proposed adjusting for covariates that may drive differential trends. Methods such as semi-parametric propensity score-based or double robust DiD estimators [1, 21] and Bayesian regression models [16] aim to reduce bias by modeling outcomes and/or weighting on observed confounders. However, their validity depends on correctly specified covariates, and unmeasured confounding in observational settings can still produce uncorrected violations. Given these limitations, sensitivity analyses have become essential tools for assessing the robustness of DiD estimates to violations of the parallel trends assumption. Rather than treating the assumption as binary, these methods formally relax it and examine how inferences shift under bounded or parameterized deviations from the ideal.

Recent advances [17, 13, 10] have introduced partial identification frameworks that yield credible bounds on treatment effects when parallel trends may not hold. These frameworks forgo point identification and instead define sets of plausible treatment effects by bounding deviations from parallel trends. For example, [17] allow for post-treatment violations constrained by smoothness (e.g., second-difference restrictions) or magnitude (e.g., bounding violations by a fixed multiple of pre-treatment differences). These non-parametric assumptions are agnostic to the violation's functional form and do not rely on stochastic modeling. Rather than estimate a violation process directly, they specify a feasible deviation set based on observed pre-trends, leading to partially identified treatment effects and valid confidence intervals.

While these methods yield robust inference, they often rely on closed-form constraints, such as bounding post-treatment deviations by pre-treatment magnitudes or enforcing smoothness, which can be overly conservative and limit their ability to capture data-driven violations. Alternatively, Bayesian approaches to sensitivity analysis may offer greater flexibility by incorporating uncertainty and prior knowledge directly into the model. For example,[11] propose an approach to relax the parallel trends assumption by modeling deviations from parallel trends as an AR(1) prior in which each period's deviation depends on the previous one and random noise. However, their model lacks a mean-shift parameter, constraining the long-run mean of deviations to zero and thus excluding systematic upward or downward trends. While [11] adopt an empirical Bayes (EB) framework, their implementations differ across applications. Their application to [3] calibrates the AR(1) parameters externally using estimates from [14], rather than estimating them directly from their own pre-treatment data. In their application to [12], they adopt a simplified random walk with drift, modeling period-to-period changes as normally distributed around a constant mean, with parameters estimated from pre-treatment data via maximum likelihood. This formulation lacks mean reversion and imposes a weaker temporal structure than AR(1), meaning parallel trends deviations can accumulate unchecked over time, thereby reducing the model's ability to distinguish temporary fluctuations from sustained shifts. While conceptually Bayesian, these models do not conduct formal sensitivity analyses—they do not systematically vary parameters or assess how treatment effect estimates change under alternative specifications.

In this study, we improve upon existing approaches by assessing treatment effects under both the parallel trends assumption and structured deviations from it, using a Bayesian framework that models and quantifies potential violations through sensitivity parameters. Focusing on the Philadelphia beverage tax (PBT) policy and using Baltimore as a control [18], we analyze beverage sales data from January 2016 to December 2017 to account for potential deviations in pre-treatment trends due to market differences and external factors. To address this, we implement Bayesian sensitivity analysis with three prior structures, including an AR(1) model for temporal deviations and an EB approach calibrated from the observed pre-treatment data. This enables a data-driven evaluation of how prior assumptions on potential deviations influence treatment effect estimates. Our results show the flexibility of Bayesian methods in handling

violations of parallel trends and highlight the importance of sensitivity analysis for robust causal inference in policy evaluation.

The remainder of the paper is organized as follows. We first formalize identification of causal effects in DiD designs under the parallel trends assumption and its violations. We then introduce a Bayesian framework for sensitivity analysis, including an AR(1) prior on deviations from parallel trends and alternative hyperparameter specification strategies, including empirical Bayes estimation. We apply this framework to the PBT to examine how posterior treatment effect estimates change under structured violations of parallel trends. In this application, we evaluate sensitivity by comparing posterior distributions of the average treatment effect on the treated (ATT) across alternative prior specifications for deviations from parallel trends, and complement this analysis with a tipping point assessment that identifies the magnitude of trend violations required to overturn policy conclusions. We conclude with a discussion of implications and extensions.

Methods

Data

We use the PBT as a policy case study to illustrate the proposed approach for evaluating policy effects in settings where pre-treatment trends may deviate from standard parallel trend assumptions. Beverage price and sales data were sourced from the market research firm Information Resources Inc (IRI), which collects data from major US retailers [15]. The PBT, implemented on January 1, 2017, imposed a 1.5 cent per ounce tax on the distribution of both sugar-sweetened and artificially sweetened beverages. This policy aimed to reduce consumption of sugary drinks and raise revenue for public programs. For this analysis, we used beverage sales data from January 1, 2016, to December 31, 2017, collected from supermarkets and pharmacies in Philadelphia and Baltimore. Observations were recorded at 26 time points, indexed as $t = 1, \ldots, 26$, where each period represents a four-week aggregation of sales. The first 13 periods ($t = 1, \ldots, 13$) correspond to the pre-tax periods in 2016, and the remaining 13 periods ($t = 14, \ldots, 26$) correspond to the post-tax periods in 2017. This structure enables consistent temporal comparisons across years. Store and beverage categorization, as well as price and sales aggregations, were conducted as described in [18].

For this study, we classified stores into two groups: supermarkets, which include grocery stores and mass merchandisers, and pharmacies, recognizing that consumer purchasing behavior may differ between these store types. In addition, a treated indicator was added to identify observations from treated locations, specifically Philadelphia during the post-tax period, defined by the indicator variable $\mathbf{1}(t > 13)$.

DiD identification under parallel trends assumption

Suppose we observe a sequence of periods $t \in \{0, 1, \ldots, T\}$, where periods $t < g$ represent pre-policy periods, and $t \geq g$ represent post-policy periods [6]. A unit is either exposed to the policy from time g onwards ($A = 1$) or remains unexposed ($A = 0$). Define the potential outcome at time t, had a unit been exposed to policy status a, as $Y^a(t)$. We are interested in estimating the ATT, which is the average difference in post-policy outcomes between exposed and unexposed units, among those exposed, expressed as:

$$\Psi_t = E[Y^1(t) - Y^0(t) \mid A = 1] \tag{1}$$

The challenge arises because we cannot directly observe $E[Y^0(t) \mid A = 1]$, as it represents the counterfactual outcome of exposed units had they not been exposed to the policy.

To address this issue, the parallel trends assumption is invoked, particularly in multi-period settings. This assumption states that, in the absence of treatment, the outcome trend among treated units would have followed the same trend as that among untreated units:

$$E[Y^0(t) - Y^0(t-1) \mid A = 1] = E[Y^0(t) - Y^0(t-1) \mid A = 0] \quad \text{for } t \geq g \tag{2}$$

This implies that we can express the counterfactual mean outcome of treated units in period t in terms of pre-policy outcomes and trends observed among untreated units:

$$E[Y^0(t) \mid A = 1] = E[Y^0(t-1) \mid A = 1] + \big(E[Y^0(t) - Y^0(t-1) \mid A = 0]\big) \tag{3}$$

Under the consistency assumption, which states that observed outcomes equal potential outcomes under the received treatment (i.e., $Y(t) = Y^a(t)$ if $A = a$), we can substitute into the ATT definition for Ψ_t:

$$\Psi_t = (E[Y(t) \mid A = 1] - E[Y(g-1) \mid A = 1]) - (E[Y(t) \mid A = 0] - E[Y(g-1) \mid A = 0]) \tag{4}$$

This equation identifies the ATT under the parallel trends assumption using observed outcome trajectories across groups. While identification only requires pre-treatment data from the immediate time before treatment ($t = g-1$), earlier pre-treatment data can still benefit estimation by improving statistical efficiency and enabling the assessment of pre-treatment trend alignment.

Violations of parallel trends assumption

The parallel trends assumption assumes that, in the absence of treatment, the outcome trends for treated and untreated groups would evolve in the same way. However, when this assumption is violated, the expected trend in the treated group may deviate from that of the untreated group. Formally, this deviation can be captured by the equation:

$$E[Y^0(t) - Y^0(t-1) \mid A = 1] = E[Y^0(t) - Y^0(t-1) \mid A = 0] + \xi_t \tag{5}$$

where ξ_t quantifies the deviation from parallel trends between time $t-1$ and t, for each $t \geq g$. If $\xi_t = 0$ for all t, then parallel trends hold, meaning that the untreated and treated groups follow the same expected trajectory. Any nonzero ξ_t introduces a deviation from parallel trends, where the sign of ξ_t determines the direction of this deviation. Given this modified parallel trends assumption, we can express the counterfactual mean outcome of the treated group at any period $t \geq g$ as:

$$E[Y^0(t) \mid A = 1] = E[Y^0(t-1) \mid A = 1] + E[Y^0(t) - Y^0(t-1) \mid A = 0] + \xi_t \tag{6}$$

This recursive equation describes how the expected counterfactual outcome at each period t is derived from its previous value, incorporating the expected change in the untreated group along with the deviation term ξ_t. To generalize this to an accumulated form, we apply this recursion iteratively from the treatment onset period g to t, summing over all incremental changes. Expanding iteratively, we obtain the telescoping sum:

$$E[Y^0(t) \mid A = 1] = E[Y^0(g-1) \mid A = 1] + \sum_{s=g}^{t} (E[Y^0(s) \mid A = 1] - E[Y^0(s-1) \mid A = 1]) \tag{7}$$

Substituting the modified parallel trends assumption into the summation, we replace each term with its equivalent expression:

$$E[Y^0(s) \mid A = 1] - E[Y^0(s-1) \mid A = 1] = E[Y^0(s) - Y^0(s-1) \mid A = 0] + \xi_s \tag{8}$$

This leads to:

$$E[Y^0(t) \mid A = 1] = E[Y^0(g-1) \mid A = 1] + \sum_{s=g}^{t} \Big(E[Y^0(s) - Y^0(s-1) \mid A = 0] + \xi_s\Big) \tag{9}$$

Applying the linearity of expectation, we simplify the summation over the untreated group's expected changes:

$$E[Y^0(t) \mid A=1] = E[Y^0(g-1) \mid A=1] + E[Y^0(t) \mid A=0] - E[Y^0(g-1) \mid A=0] + \sum_{s=g}^{t} \xi_s \tag{10}$$

Finally, the consistency assumption allows us to replace remaining potential outcomes with observed outcomes. We equate the treated and control group's potential outcomes at time t with their observed outcomes, replacing $E[Y^1(t) \mid A=1]$, $E[Y^0(t) \mid A=0]$, and $E[Y^0(g-1) \mid A=0]$ with their observed values. Under the arrow of time or no anticipation assumption, future intervention does not affect past outcomes, allowing us to equate $E[Y^0(g-1) \mid A=1]$ with $E[Y(g-1) \mid A=1]$, yielding:

$$E[Y(t) \mid A=1] = E[Y(g-1) \mid A=1] + E[Y(t) \mid A=0] - E[Y(g-1) \mid A=0] + \sum_{s=g}^{t} \xi_s \tag{11}$$

This final equation captures the cumulative deviation of the treated group's counterfactual outcomes from the untreated group's expected trend, accounting for both structural trend differences and accumulated deviations ξ_s. The summation term reflects how deviations at each time s accumulate, leading to a total shift in expectations by time t.

Two-way fixed effects estimation

We review the two-way fixed effects (TWFE) framework as the baseline approach for estimating treatment effects under the parallel trends assumption. Each expectation above can be estimated using a multi-period TWFE model, which is specified as follows:

$$E[Y_i(t) \mid A=a] = \theta_t + \gamma + \beta \mathbf{1}_{\{t \geq g, A=1\}} \tag{12}$$

In this model, $Y_i(t)$ represents the outcome variable for unit i at time t. The term θ_t captures time-specific fixed effects, which account for any time-varying factors common across all units, effectively controlling for temporal trends unrelated to the policy. The term γ represents group-specific fixed effects, which account for unobserved characteristics specific to treated ($A=1$) or untreated ($A=0$) groups that remain constant over time. The parameter β is associated with the time-dependent treatment status indicator $\mathbf{1}_{\{t \geq g, A=1\}}$, and it represents the treatment effect for exposed units in periods $t \geq g$, which corresponds to the ATT in this multi-period setting. Posterior inference for Ψ_t can then be achieved by estimating the posterior distribution of β using a Bayesian linear regression model.

AR(1) prior on deviations from parallel trends

To allow for deviations from the parallel trends assumption, we introduce an AR(1) prior on the sequence of post-treatment violation terms $\{\xi_g, \xi_{g+1}, \ldots, \xi_t\}$. The AR(1) prior provides a structured way to model how deviations might evolve over time, ensuring that each period's violation builds incrementally upon the prior period's deviation. Specifically, for each period $s = g, g+1, \ldots, t$, we define:

$$\xi_s = \eta(1-\rho) + \rho\, \xi_{s-1} + \sigma\, \varepsilon_s \tag{13}$$

Here, $\varepsilon_s \sim$ iid $N(0,1)$, and η, ρ, and σ are the parameters of the process. This structure is denoted as $\{\xi_s\}_{s=g}^{t} \sim gAR1(\eta, \rho, \sigma)$. The mean of this AR(1) is $E[\xi_s] = \eta$. The variance is given by $V[\xi_s] = \frac{\sigma^2}{1-\rho^2}$. The parameter $-1 < \rho < 1$ is the autoregressive parameter that governs the dependence on the state, and σ is the standard deviation of the noise term. Under this specification, each deviation ξ_s is incrementally dependent on the deviation of the preceding period ξ_{s-1}. Consequently, the cumulative deviation term for periods g to t given by $\sum_{s=g}^{t} \xi_s$ captures structured incremental departures at each step,

as each deviation systematically builds on the previous one. This AR(1) prior offers several advantages for modeling longitudinal deviations in DiD settings. By explicitly accounting for autocorrelation, it ensures that deviations are not treated as independent across time but evolve gradually, reflecting realistic temporal dependencies common in panel or repeated-measures data. This not only enables more robust inference in the presence of serially correlated violations but also regularizes the trajectory of ξ_s, preventing overfitting to local noise.

Our approach is related but differs from [11], who model the deviation process as an autoregressive structure with no mean shift. In their formulation, the deviation term (analogous to our ξ_s) follows $\xi_s = \rho\xi_{s-1} + \sigma\varepsilon_s$, thereby assuming the deviations are centered around zero over time. This corresponds to an AR(1) process with zero long-run mean. In contrast, we explicitly incorporate a mean-shift parameter η in the AR(1) formulation, $\xi_s = \eta(1-\rho) + \rho\xi_{s-1} + \sigma\varepsilon_s$, allowing for persistent directional shifts in the counterfactual trend. Moreover, rather than calibrating parameters externally as done in their application, we estimate (η, ρ, σ) empirically from pre-treatment data, yielding a fully data-adaptive prior. Key notation is summarized in Table 1.

Symbol	**Description**
θ_t	Time-specific effect at period t in the DiD model
γ	Group-specific effect (e.g., $a = 1$ for treatment, $a = 0$ for control)
β	Treatment effect for periods $t \geq g$ in the TWFE model
A	Group indicator ($A = 1$ for treated, $A = 0$ for control)
t	Time period indicator ($t \in \{0, 1, \ldots, T\}$)
g	Treatment onset time (the first period when $t \geq g$)
s	Index for summing over time periods from g to t
Ψ_t	Average Treatment Effect on the Treated at period t
ξ_t	Sensitivity parameter for parallel trend violation at period t
η	Long-term mean parameter in the AR prior
ρ	Autoregressive parameter in the AR prior
σ	Standard deviation of the noise term in the AR prior

Table 1: Notation reference for Bayesian DiD model

Hyperparameter specification strategies

In practice, inference under violations of the parallel trends assumption depends on how the deviation process is specified. To assess how sensitivity analyses vary under different prior assumptions, we consider three strategies for specifying the hyperparameters of the AR(1) deviation process, namely the standard deviation σ, autoregressive coefficient ρ, and long-run mean η. Specifically, we compare models with fixed hyperparameter values (Fixed-1, Fixed-2, Fixed-3), models with fully Bayesian hyperparameter priors (Fully-1, Fully-2, Fully-3), and models using empirical Bayes hyperparameter estimates derived from pre-treatment data (EB-1, EB-2, EB-3). Each specification is applied separately to the supermarket and pharmacy datasets, with full parameter configurations listed in Table 2.

In the fixed models, we fix the autoregressive parameter at $\rho = 0.95$, a value close to 1 that preserves strong temporal dependence while still allowing the AR(1) to include a non-zero long-run mean η. We draw $\eta \sim \text{Uniform}(0.1, 0.9)$ and vary the standard deviation σ across 0.001 (Fixed-1), 1 (Fixed-2), and 5 (Fixed-3) to represent increasingly flexible assumptions about the magnitude of deviations from parallel trends. A very small σ tightly constrains deviations toward the mean, approximating the parallel trends assumption, while larger values permit increasingly flexible departures. This range allows us to examine how posterior treatment effect estimates change under increasingly permissive deviation specifications. Because these hyperparameters are fixed rather than estimated, this setup simplifies the sensitivity analysis by isolating the role of deviation scale, but it omits posterior uncertainty in the autoregressive structure and may overstate confidence as it does not incorporate uncertainty in ρ and σ.

AR(1) models	Supermarket			Pharmacy		
	σ	ρ	η	σ	ρ	η
Fixed-1	0.001	0.95	U(0.1, 0.9)	0.001	0.95	U(0.1, 0.9)
Fixed-2	1	0.95	U(0.1, 0.9)	1	0.95	U(0.1, 0.9)
Fixed-3	5	0.95	U(0.1, 0.9)	5	0.95	U(0.1, 0.9)
Fully-1	HN(1)	B(2,2)	U(0.1, 0.9)	HN(1)	B(2,2)	U(0.1, 0.9)
Fully-2	HN(2)	B(2,2)	U(0.1, 0.9)	HN(2)	B(2,2)	U(0.1, 0.9)
Fully-3	HN(5)	B(2,2)	U(0.1, 0.9)	HN(5)	B(2,2)	U(0.1, 0.9)
EB-1	0.166	0.371	1.60	0.340	0.785	1.64
EB-2	0.166	0.742	1.60	0.340	1.57	1.64
EB-3	0.166	1.113	1.60	0.340	2.36	1.64

Table 2: Configurations of AR(1) models with parameter configurations across supermarket and pharmacy data. U = Uniform, HN(k) = HalfNormal(k), B = Beta.

The fully Bayesian models incorporate fully Bayesian priors to account for uncertainty in the autoregressive structure. We place a Beta$(2, 2)$ prior on ρ, reflecting moderate belief in positive autocorrelation while avoiding extremes near 0 or 1. For the noise scale, we assign $\sigma \sim \text{HalfNormal}(k)$ with $k \in \{1, 2, 5\}$ across the models Fully-1 to Fully-3, respectively, allowing increasing flexibility in deviation magnitude. The long-run mean η remains drawn from Uniform$(0.1, 0.9)$. This approach enables the model to learn both the smoothness and volatility of deviations from the data, producing more realistic posterior uncertainty. However, its effectiveness may still depend on the informativeness of the priors, especially when the number of post-treatment periods is short.

The EB models estimate ρ, σ, and η from pre-treatment trends using an EB procedure (see next subsection). EB-1 uses the raw EB estimates, while EB-2 and EB-3 apply multiplicative scaling to σ by factors of 2 and 5, respectively, to evaluate robustness to the assumed magnitude of deviations from parallel trends. This data-driven approach allows the prior to adapt to observed trends, but may be sensitive to noise or bias in the pre-treatment period.

EB estimation of AR(1) parameters

To enable data-adaptive regularization, we empirically estimate parameters for the AR(1) model (13) directly from observed pre-treatment trends. Let X_t denote the sequence of estimated violation terms ξ_t over the pre-treatment periods. We rewrite the model as $X_t = c + \rho X_{t-1} + \sigma\epsilon_t$, where $\epsilon_t \sim N(0, 1)$, and $c = \eta(1 - \rho)$. This representation makes the model linear in c and ρ. Using ordinary least squares, we estimate

$$\begin{bmatrix} c \\ \rho \end{bmatrix} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{X} \tag{14}$$

where $\mathbf{Z}$ includes a column of ones and lagged values X_{t-1}. This procedure minimizes the residual sum of squares, yielding an estimate of ρ as the influence of X_{t-1} on X_t. With estimates for c and ρ, we derive η as $\eta = \frac{c}{1-\rho}$, providing an estimate of the long-term mean based on the short-term dynamics captured by the model. To estimate σ, which represents the standard deviation of the noise term, we use the residuals from the model, calculated as $\hat{\epsilon}_t = X_t - (c + \rho X_{t-1})$. These residuals approximate the noise component $\sigma\epsilon_t$ in (13). The empirical variance of these residuals provides an estimate for σ^2:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{t=2}^{n} \left(X_t - (c + \rho X_{t-1})\right)^2 \tag{15}$$

Taking the square root of $\hat{\sigma}^2$ gives $\hat{\sigma}$, which serves as an empirical measure of the variability captured by σ in (13).

Implementation setup

The models were implemented in PyMC (version 4.0) [20]. The default sampler is the no-U-turn sampler, an adaptive form of Hamiltonian Monte Carlo [4, 9]. The default number of chains is four, and the default runs 1000 warmup iterations (for burn-in and adaptation) and 1000 sampling iterations. All sampling runs ended with split-$\widehat{R}$ values less than 1.01 for all parameters, indicating consistency with convergence to approximate stationarity [7].

Results

To evaluate how assumptions about violations of the parallel trends assumption influence treatment effect estimates, we perform a series of Bayesian sensitivity analyses using the AR(1) prior configurations described in Table 2. While nine total model configurations were used across supermarket and pharmacy datasets, we focus on Fixed-1, Fully-1, and EB-1 models as representative trends from each hyperparameter strategy in Table 2. The full results for all models are provided in Figure 2.

Counterfactual trends and ATT estimates under parallel trends violations

Each panel in Figure 1 illustrates how alternative assumptions about counterfactual trends translate into different estimates of the beverage tax effect, shown through two components: observed and counterfactual trends (left) and the ATT posterior (right). The left panels show observed sales for treatment and control groups, alongside two modeled counterfactuals: one based on extrapolating pre-treatment trends under the parallel trends assumption, and another adjusted by an autoregressive deviation process governed by a sensitivity parameter ξ, which allows for flexible violations of the parallel trends assumption. The ATT posterior histogram summarizes the estimated treatment effects under each assumption, calculated as the difference between observed treatment outcomes and their corresponding counterfactual outcomes.

In the supermarket panels of Figure 1, the Fixed-1, Fully-1, and EB-1 models illustrate how different specifications of the deviation process alter the estimated counterfactual trend for Philadelphia in the absence of the tax. Under the Fixed-1 model, the counterfactual trend that allows deviations from parallel trends shifts gradually downward relative to the parallel-trend extrapolation, reflecting the accumulation of modest but persistent deviations over time. The Fully-1 model shows greater divergence and wider uncertainty because its priors place fewer constraints on the deviation process. In particular, the priors $\rho \sim \text{Beta}(2, 2)$ and $\sigma \sim \text{HN}(1)$ place relatively weak constraints on the deviation process, allowing the counterfactual trend to depart from the parallel-trend extrapolation when the posterior permits persistent deviations from parallel trends that could explain the observed divergence after treatment. In contrast, the EB-1 model remains closely aligned with the extrapolated parallel-trend counterfactual because its empirically estimated parameters imply near-stationarity and relatively small cumulative deviations. These differences in counterfactual trend behavior translate directly into differences in the posterior distributions of the ATT. The Fully-1 model produces the most attenuated treatment effect estimates under violations of the parallel trends assumption, whereas EB-1 yields the most stable estimates. Similar patterns are observed in the pharmacy dataset. Taken together, these results indicate that the magnitude of the estimated PBT effect depends on how strongly the model constrains counterfactual sales trends to follow observed pre-treatment behavior.

Sensitivity of ATT estimates to hyperparameter specifications

Figure 2 summarizes the posterior mean ATT (μ) and the corresponding 95% credible intervals across all nine AR(1) hyperparameter configurations defined in Table 2. These estimates are derived from the ATT

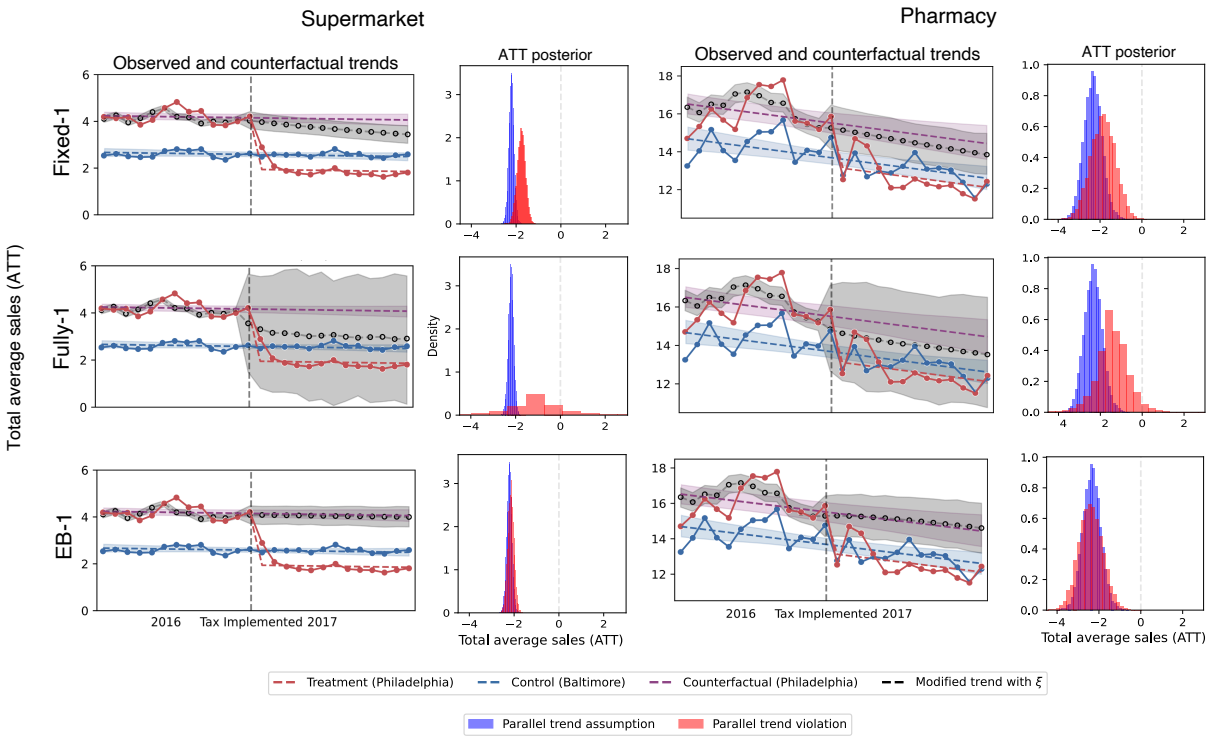

Figure 1: Observed and counterfactual trends and ATT posterior for the Supermarket and Pharmacy datasets. The observed and counterfactual trends plots (left panels) show the observed treatment trend (Philadelphia), observed control trend (Baltimore), counterfactual trend under the parallel trends assumption, and counterfactual trend incorporating an AR(1) deviation process with sensitivity parameter ξ. The ATT posterior plots (right panels) show the posterior distribution of the treatment effect, defined as the difference between observed treatment outcomes and their corresponding counterfactual outcomes. The histograms reflect the posterior distribution of the total ATT accumulated over the post-treatment period, up to the final time point. Each row corresponds to Fixed-1, Fully-1, and EB-1 models from top to bottom respectively, as defined in Table 2. The analysis highlights the impact of increasing sensitivity parameters on outcome trends and ATT posteriors.

posterior distributions illustrated in Figure 1. While Figure 1 displays the full posterior distributions for representative model specifications, Figure 2 provides a concise numerical summary across all model configurations. Specifically, the left panels summarize ATT estimates implied by the modified counterfactual trends incorporating the AR(1) deviation process with sensitivity parameter ξ, corresponding to the "Modified trend with ξ" line in the observed and counterfactual trends plots in Figure 1. The right panels summarize ATT estimates under the parallel trends violation assumption, defined as the difference between observed treatment outcomes and their counterfactual outcomes. These correspond to the "Parallel trend violation" distributions in the ATT posterior plots in Figure 1. The parameter μ denotes the posterior mean of the cumulative ATT over the post-treatment period implied by the counterfactual trends generated under each AR(1) specification. Increasing model flexibility through higher σ (Fixed-1 to Fixed-3), wider priors (Fully-1 to Fully-3), or EB scaling (EB-1 to EB-3) leads to greater posterior uncertainty and shifts in the mean estimates. The fixed models show a clear gradient: as σ increases, so does the variance in the counterfactual outcome trends and the resulting ATT estimates. The fully Bayesian models mirror this, with HalfNormal priors capturing uncertainty more realistically but still

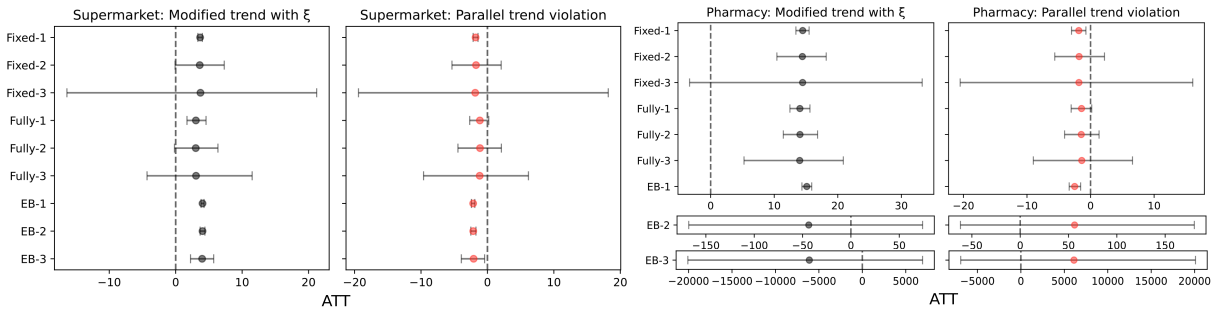

Figure 2: Posterior mean ATT (μ) and 95% credible intervals for ATT estimates across all AR(1) configurations from Table 2. Each point represents the posterior mean of the cumulative ATT over the post-treatment period, with error bars indicating the corresponding 95% credible interval. These summaries are derived from the ATT posterior distributions shown in Figure 1. The left panels summarize ATT estimates implied by the modified counterfactual trends incorporating the AR(1) deviation process with sensitivity parameter ξ, corresponding to the "Modified trend with ξ" line in the observed and counterfactual trends plots in Figure 1. The right panels summarize ATT estimates under the parallel trends violation assumption, corresponding to the "Parallel trend violation" distributions in the ATT posterior plots in Figure 1. For the pharmacy dataset, EB-2 and EB-3 are plotted on separate horizontal scales due to extreme estimates driven by nonstationary ρ values ($|\rho| \geq 1$), which amplify deviations and inflate credible intervals.

exhibiting sensitivity to prior scale. In the EB models, posterior sensitivity is generally low, especially in EB-1, where all AR(1) parameters are directly calibrated from pre-treatment data. However, EB-2 and especially EB-3 yield unstable results in the Pharmacy dataset due to excessively high estimated values of ρ (1.57 and 2.36, respectively). Since AR(1) processes with $|\rho| \geq 1$ become nonstationary, these cases amplify small fluctuations, leading to extreme values and inflated credible intervals. In practice, posterior estimates with $|\rho| \geq 1$ should be treated as a diagnostic signal that the model may be overfitting short-term fluctuations. To prevent nonstationary drift that obscures attribution of post-tax sales changes to the policy itself, it is important to constrain persistence in counterfactual trends, for example by using priors such as Beta distributions over $\rho \in (0, 1)$ that enforce or softly encourage stationarity, especially when pre-treatment data are limited.

Tipping point analysis for sensitivity to parallel trends violations

To assess the robustness of our treatment effect estimates to violations of the parallel trends assumption, we examine how large the sensitivity parameter η, which governs the magnitude of deviations from the parallel trend in the AR(1) process, must be before the upper bound of the ATT's 95% CI includes zero. Since $\eta = 0$ corresponds to strict parallel trends, we vary η across a range of values, focusing primarily on negative values that shift the ATT toward the null, to test how easily the effect could be attenuated. Figure 3 shows posterior mean ATT (μ) and corresponding 95% credible intervals across values of η under Fixed-1, Fully-1, and EB-1 models. In our AR(1) framework, as η introduces additive shifts in sales growth rates, exponentiating these deviations yields multiplicative changes in sales volume, enabling fold-change interpretation relative to the parallel trend.

In our default specifications, the supermarket and pharmacy models assume $\eta = 1.6$ and $\eta = 1.64$, respectively (Table 2). Based on the PBT dataset [18], average beverage sales were 4.85 million ounces per 4-week period for supermarkets and 0.16 million ounces for pharmacies. For supermarkets under Fully-1 model, the tipping point occurs at $\eta = -0.46$, implying that Philadelphia sales under the counterfactual without the tax would have had to increase by approximately 58% more than Baltimore sales. Applying this increase to the original average of 4.85 million ounces yields an additional 2.81 million ounces per store. Since one can of soda is 12 ounces, this corresponds to roughly 234,000 extra cans per 4-week period.

For pharmacies under Fully-1 model, the tipping point at $\eta = -0.35$ implies a 42% increase in untreated sales. When applied to the baseline volume of 0.16 million ounces, this equates to approximately 67,000 more ounces, or about 5,600 additional cans. EB-1 model reaches its tipping point at $\eta = -1.2$, requiring a 3.32-fold increase in sales. This equates to 11.25 million extra ounces (938,000 cans) for supermarkets and 371,000 ounces (30,900 cans) for pharmacies. The Fixed-1 model reaches their tipping points at $\eta = -6.89$ and $\eta = -4.64$, requiring 980-fold and 104-fold increases, which is over 4.7 billion additional ounces (395 million cans) for supermarkets and 16.4 million ounces (1.37 million cans) for pharmacies. These results show that the Fixed-1 model demands large counterfactual increases to overturn the effect, the EB-1 model requires substantial but unlikely deviations, and the Fully-1 model is the most sensitive to modest departures from parallel trends.

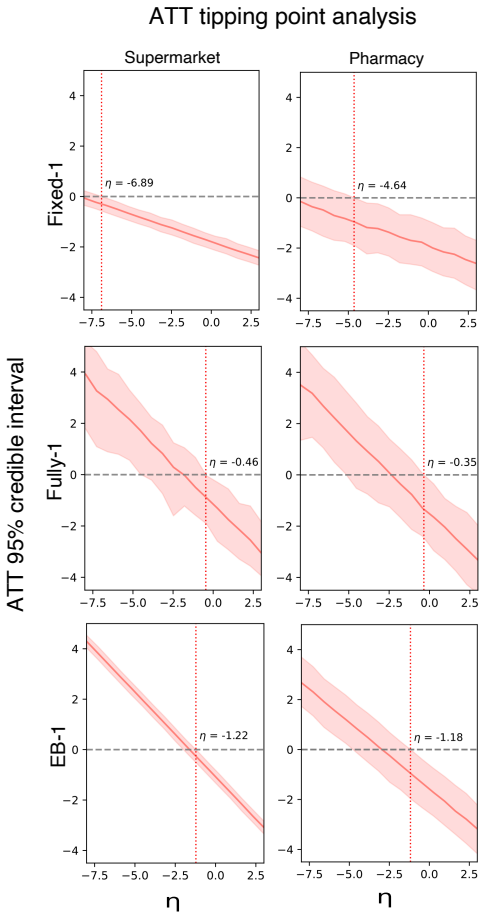

Figure 3: Tipping point analysis assessing the robustness of ATT estimates to violations of parallel trends. Each panel shows the posterior μ and 95% CI of the ATT across values of the sensitivity parameter η, which controls the degree of deviation via an AR(1) structure. The dotted red line marks the η value where the upper bound of the 95% CI crosses zero—indicating the tipping point beyond which the ATT is no longer significantly negative. Results are shown for three representative models (Fixed-1, Fully-2, and EB-1) across supermarket (left) and pharmacy (right) datasets.

Discussion

Our Bayesian sensitivity framework extends DiD analyses by explicitly modeling potential violations of the parallel trends assumption through an autoregressive deviation process. By introducing an AR(1) prior on deviations from the counterfactual trend and incorporating a mean shift parameter that captures systematic departures from parallel trends, the approach provides a structured way to assess how treatment effect estimates depend on assumptions about counterfactual dynamics. The framework further enables tipping point analyses that quantify how large deviations from parallel trends must be to overturn the estimated treatment effect and allows researchers to examine how alternative prior specifications influence sensitivity analyses. Applied to Philadelphia's sweetened beverage tax, this approach illustrates how policy effect estimates depend on assumptions about counterfactual parallel trends, which we formalize through a sensitivity parameter governing the magnitude of trend deviations.

Across both supermarket and pharmacy datasets, the different sensitivity configurations imply different interpretations of the policy effect when parallel trends are violated. Under the Fixed configurations, increasing the assumed magnitude of trend violations directly inflates posterior uncertainty in the ATT, with larger deviations producing wider credible intervals and more divergence in the modified counterfactual outcome trends. This provides a transparent assessment of sensitivity, but relies on fixed values governing how violations accumulate over time. The fully Bayesian configurations yield a more cautious interpretation by allowing greater flexibility in how deviations evolve over time. As uncertainty in the deviation process increases, counterfactual outcome trends become more dispersed and posterior ATT estimates widen or attenuate toward the null, reflecting how weaker constraints on trend violations propagate into less precise policy effect estimates. The EB configurations support robust policy inference when pre-treatment trends are informative, yielding counterfactual trajectories that remain close to the parallel-trend extrapolation and ATT estimates that are stable across assumptions. However, when empirical calibration implies highly persistent deviations over time ($|\rho| \geq 1$), the resulting counterfactual trends can become nonstationary and posterior estimates extreme. In such cases, instability reflects the accumulation of implausibly large trend deviations rather than meaningful evidence about the policy effect. Together, these results illustrate how modeling trend violations enables policymakers to distinguish between sensitivity driven by implausible assumptions and uncertainty grounded in observed data.

To further clarify whether conclusions about the beverage tax hinge on realistic or implausible assumptions about unobserved trends, we examine a tipping point analysis. For Fixed configurations, overturning the estimated effect would require counterfactual changes in sales that far exceed observed pre-treatment variability. Fully Bayesian configurations reach the tipping point under more moderate departures, indicating greater caution when counterfactual dynamics are weakly constrained. EB configurations provide a balanced assessment, yielding stable conclusions when pre-treatment trends are informative while clearly signaling when policy inference becomes unreliable under implausible nonstationary assumptions. In practice, this suggests that empirically calibrated priors may offer a pragmatic compromise between fixed specifications that may overstate robustness and fully Bayesian specifications that can become highly sensitive when the deviation process is weakly constrained.

While these analyses clarify the robustness of the estimated policy effects, important limitations warrant consideration. Our empirical analysis relies on a two-region comparison between Philadelphia and Baltimore, which does not capture potential spillover effects or cross-border shopping behavior that may influence nearby regions. Expanding the analysis to include indirectly affected regions and covariates for regional heterogeneity could offer a more complete view of the policy's broader impact, though this introduces additional complexity within a two-way fixed effects framework [21, 8]. The EB approach also depends on the stability and informativeness of pre-treatment trends. When pre-policy periods are short or subject to structural change, empirical calibration may impose misleading regularization, reducing the reliability of resulting inferences.

Overall, our study suggests that credible inferences about policy effects remain possible even when parallel trends are violated, with sensitivity analyses offering a flexible framework that allows practitioners to gauge how conclusions vary with different modeling choices and pre-treatment information.

Author contributions

S.W.H. wrote the main manuscript text, conducted analyses, and prepared analyses. All authors conceptualized the study design and key methods, interpreted results, and reviewed and revised the manuscript.

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Data availability

The price and sales data used in this study were obtained under license from Information Resources Inc. (IRI) and are subject to contractual restrictions that prohibit public sharing.

Conflict of interest

The authors declare that they have no conflict of interest.

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